

1) Find the volume generated by rotating the region bounded by the curves around the y -axis. The curves are $y = \frac{1}{x}$, $y = 0$, $x = 1$, and $x = 4$. Use cylindrical shells.

2) Find the volume generated by rotating the region bounded by the curves around the y -axis. The curves are $y = x^3 + x$, $y = 0$, $x = 0$, and $x = 1$. Use cylindrical shells.

3) Consider the region bounded by the x -axis and the equation $y = \sqrt{r^2 - x^2}$. Rotate this region about the x -axis and identify the shape created. Then use either the shells method or the disk/washers method find the volume and compare to common mathematical knowledge. Decide which technique is more appropriate. (Hint: you know this shape already, just get there conceptually and then let the calculus catch up to you.)

4) Find the volume generated by wrapping the area bounded by $y = x^2$ and $y = 4x - x^2$ about the line $x = -3$ using either shells or disks/washers.

5) Find the arc length of the curve $y = -4x + 1$ from $x = -1$ to $x = 1$ by first using the Pythagorean Theorem and then by using integration. Your answers should agree.

6) Use calculus to find the arc length of the curve $y = \sqrt{4 - x^2}$ from $x = -2$ to $x = 2$.

7) Set up (but do not solve) the integral that would give the surface area of a solid created by rotating $y = \sqrt{x^2 - 2}$ around the x -axis from $x = 2$ to $x = 6$. Once your integral is set up, approximate its value by using technology.

8) Set up (but do not solve) the integral that would give the surface area of a solid created by rotating $x = \sin(y)$ around the y -axis from $y = 0$ to $y = \pi$. Once your integral is set up, approximate its value by using technology.

9) Mixed review time! Compute **both** the integral and derivative of each of the following four functions. For derivatives, product and chain rules will be needed. For integrals you will only need u -substitution techniques and basic facts.

$$f(x) = e^{3x}$$

$$g(x) = xe^{3x^2}$$

$$h(x) = x^2e^{3x^3}$$

$$j(x) = \sin(x)e^{\cos(x)}$$

10) Mixed review time! Compute the following integral by setting $u = x^2 + 1$. Please note that this will be one of (if not) the last times I will provide the choice for u . If you are still struggling to see the optimal choice please reach out so we can drill them for twenty minutes or until you feel more confident.

$$\int \frac{x^3}{x^2 + 1} dx$$